A lower bound for matrix multiplication

$R_K\langle 3,3,3\rangle \geqslant 21$

Mathematical proof notes

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Main theorem. *Let K be an arbitrary field. If the identity*

$$BA = \sum_{i=1}^{r} a_i(A)b_i(B)W_i$$

holds for all matrices $A, B \in \mathrm{Mat}_3(K)$, where a_i, b_i are linear forms and the matrices $W_i \in \mathrm{Mat}_3(K)$ are fixed, then $r \geqslant 21$.

Contents

1 Matrix notation and basic identities **2**

2 Coordinatewise products and conductors **3**

2.1 Connectedness and coordinate components 4

3 Normalizing the algorithm and dimension bounds **5**

3.1 Choosing a hyperplane 5

3.2 Matrix representatives of the conductors 6

3.3 Joint symmetrizers 8

4 Supports of tests and atomic tests **10**

4.1 Two bounds for tests with prescribed support 10

4.2 Atomic tests 12

5 Excluding a positive dimension increment **13**

5.1 Local geometry of a growing component 13

5.2 Replacing one bilinear pair 14

6 Four atoms: a residual sum of eight terms **16**

6.1 A five-dimensional annihilator 16

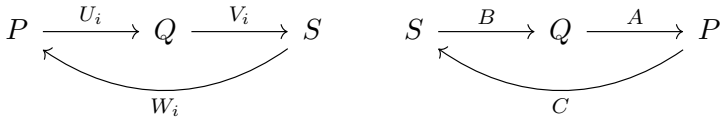
6.2 Coordinates of a three-dimensional kernel 16

6.3 Quadratic separation and concurrent lines 17

7 Three atoms and completion of the proof **20**

1 Matrix notation and basic identities

All vectors in P, Q, S are regarded as row vectors; matrices act on the right. For now we allow dimensions $p = \dim P$, $q = \dim Q$, $s = \dim S$, where $p, s \geqslant 1$ and $q \geqslant 2$. The coefficient matrices and inputs act in the following directions:



For example, U_i has size $p \times q$, A has size $q \times p$, and BA has size $s \times p$. In a product UV, U acts first and V second. A column η defines the linear form $x \mapsto x\eta$ on row vectors. If a is a nonzero column and b a nonzero row, the ordinary matrix product ab has rank one. We call the lines Ka and Kb its *column direction* and *row direction*, respectively. The *column support* of a family of matrices is the span of all their columns; its *row support* is the span of all their rows, equivalently the sum of their images under the right action.

Definition 1.1. A *bilinear algorithm of length* r consists of r triples of matrices (U_i, V_i, W_i), indexed by a set I, satisfying

$$BA = \sum_{i \in I} \operatorname{tr}(U_i A) \operatorname{tr}(V_i B) W_i \qquad \text{for all } A, B. \tag{1.1}$$

All algorithms considered here are exact: the identity holds over the field itself, with no limiting process. We omit this qualification below. The minimum length for this format is denoted by $R_K\langle p, q, s\rangle$.

Every linear form on matrices is uniquely of the form $A \mapsto \operatorname{tr}(UA)$: its value on the matrix unit E_{ab} is U_{ba}. Thus this definition agrees with the formulation of the main theorem.

Definition 1.2. A matrix $C : P \to S$, representing the output linear form $BA \mapsto \operatorname{tr}(CBA)$, will be called a *test*. The *maps defined by the coefficient forms* are denoted by

$$\alpha(A)_i = \operatorname{tr}(U_i A), \quad \beta(B)_i = \operatorname{tr}(V_i B), \quad \gamma(C)_i = \operatorname{tr}(W_i C).$$

They take values in K^I. On this space we use the coordinatewise product $xy = (x_i y_i)_i$, the identity $\mathbf{1} = (1)_i$, and the pairing $\langle x, y\rangle = \sum_i x_i y_i$. For subspaces $X, Y \subseteq K^I$, XY denotes the *linear span* of all products xy. The support of a vector x is $\operatorname{supp} x = \{i : x_i \neq 0\}$; the support of a subspace is the union of the supports of its vectors.

With respect to a nondegenerate pairing, the *annihilator* of a subspace consists of all elements of the other space that pair to zero with it. For matrices we use $\operatorname{tr}(XY)$, and for coordinate vectors we use $\langle x, y\rangle$. In the latter case we also write $X^\perp$ and speak of orthogonality; no positive definiteness is assumed.

Lemma 1.3. *The maps α, β, γ are injective. For all matrices of the appropriate sizes, the following identities hold:*

$$\operatorname{tr}(CBA) = \sum_i \alpha(A)_i \beta(B)_i \gamma(C)_i, \tag{1.2}$$

$$CB = \sum_i \beta(B)_i \gamma(C)_i U_i, \qquad AC = \sum_i \gamma(C)_i \alpha(A)_i V_i, \tag{1.3}$$

$$\sum_i \gamma(C)_i U_i X V_i = \operatorname{tr}(X) C \qquad (X \in \operatorname{Mat}_q(K)), \tag{1.4}$$

$$\sum_i \operatorname{tr}(X V_i T U_i) W_i = \operatorname{tr}(X) T \qquad (T \in \operatorname{Mat}_{s,p}(K)). \tag{1.5}$$

Proof. The first identity follows by multiplying (1.1) by C and taking the trace. Nondegeneracy of $\operatorname{tr}(XY)$, checked on matrix units, gives both identities in (1.3). If, for example, $\alpha(A) = 0$, then $BA = 0$ for every B, hence $A = 0$. Cyclic permutation gives the other two injectivity statements.

To prove (1.4), first substitute matrix units for A, B in (1.2). Comparing coefficients gives

$$\sum_i (U_i)_{ab}(V_i)_{cd}\gamma(C)_i = \delta_{bc}C_{ad}.$$

Multiplying by X_{bc} and summing yields (1.4). Trace duality in C gives (1.5). All these identities are polynomial in the coefficients and remain valid after a field extension. □

Zero summands may be deleted. Assuming an algorithm of length at most twenty exists, pass to an algebraic closure of K and delete its zero summands. Unless stated otherwise, from this point until the final section, K is algebraically closed, all U_i, V_i, W_i are nonzero, and $r \leqslant 20$.

2 Coordinatewise products and conductors

Definition 2.1. For subspaces $M, N \subseteq K^I$, their *conductor* is

$$(N : M) = \{d \in K^I : md \in N \text{ for every } m \in M\}.$$

This is a subspace of K^I; the colon does not denote a quotient space. The *multiplicative stabilizer* of X is the algebra $(X : X) = \{a : aX \subseteq X\}$. Its elements need not be invertible.

Lemma 2.2. *Let $X, Y \subseteq K^I$ be nonzero subspaces such that $xy = 0$ with $x \in X$, $y \in Y$ implies $x = 0$ or $y = 0$. Then*

$$\dim XY \geqslant \dim X + \dim Y - 1.$$

Proof. Induct on $\dim X$. If $\dim X = 1$, multiplication by any nonzero $x \in X$ is injective on Y. Otherwise choose a coordinate i on which some product is nonzero. The spaces $X_0 = \{x \in X : x_i = 0\}$ and $(XY)_0 = \{v \in XY : v_i = 0\}$ have codimension one in X and XY, respectively, and $X_0Y \subseteq (XY)_0$. The induction hypothesis applied to X_0, Y gives the required bound. □

Definition 2.3. For $G \subseteq I$, set

$$X[G] = \{x \in X : x_i = 0 \text{ if } i \notin G\}, \qquad X|_G = \{(x_i)_{i\in G} : x \in X\}.$$

These are the *coordinate section* and the *restriction* of the space, respectively. Their dimensions need not be equal.

Lemma 2.4. *Let $M = \operatorname{span}(\mathbf{1}, x, y)$ and $N, B \subseteq K^I$, with $MB \perp N$. Then*

$$\dim(N : M) + \dim(B : M) \geqslant 3\dim N + 3\dim B - 2r.$$

Proof. For any $L \subseteq K^I$, consider

$$\Phi_L : L^3 \longrightarrow K^I \oplus K^I, \qquad (l_0, l_1, l_2) \longmapsto (l_1 - xl_0, l_2 - yl_0).$$

Its kernel consists of the triples (l_0, xl_0, yl_0) with $l_0 \in (L : M)$. Thus

$$\dim \operatorname{im} \Phi_L = 3\dim L - \dim(L : M).$$

The form on $K^I \oplus K^I$ defined by

$$\omega((a,b),(c,d)) = \langle a,d\rangle - \langle b,c\rangle$$

is nondegenerate and alternating, meaning $\omega(v,v) = 0$, including in characteristic two. The images of Φ_N and Φ_B are orthogonal: on expanding, the terms with factors $1, x, y$ vanish by hypothesis, while the terms with xy cancel. The dimensions of orthogonal subspaces sum to at most $2r$. Substitute the dimensions computed above. □

2.1 Connectedness and coordinate components

Definition 2.5. A finite nonempty family of nonzero vectors $(v_i)_{i\in G}$ is *connected* if every partition $G = G_1 \sqcup G_2$ into two nonempty parts satisfies

$$\operatorname{span}\{v_i : i \in G_1\} \cap \operatorname{span}\{v_i : i \in G_2\} \neq 0.$$

By splitting disconnected families, we obtain the unique finest partition for which the span of the family is the direct sum of the spans of the parts. These parts are called its connected components.

Uniqueness also follows from a description by minimal linear dependences: all indices in one such dependence must belong to the same part; the transitive closure of these relations gives the components.

Lemma 2.6. *Suppose that the nonzero rank-one matrices a_ib_i form a connected family. Then*

$$\dim \operatorname{span}\{a_i\} + \dim \operatorname{span}\{b_i\} \leqslant \dim \operatorname{span}\{a_ib_i\} + 1.$$

Proof. Start with one matrix. Suppose the inequality has been proved for the part J already chosen. If $J \neq G$, connectedness allows us to choose a minimal nonempty part $H \subseteq G \setminus J$ whose span intersects the span of J nontrivially. The matrices indexed by H are independent; the intersection is one-dimensional and spanned by $T = \sum_{i\in H} c_i a_i b_i \neq 0$, where every $c_i \neq 0$. Otherwise one could remove an index from H.

Write A_0, B_0 for the previous spans of the factors and A_1, B_1 for the enlarged spans. The columns of T lie in A_0, and its rows lie in B_0. This implies

$$(\dim A_1 - \dim A_0) + (\dim B_1 - \dim B_0) \leqslant |H| - 1.$$

Here is a direct linear algebra verification of this last step. Factor $T = XY$, with columns c_ia_i in X and rows b_i in Y. Projecting the columns modulo A_0 gives $\bar{X}Y = 0$; projecting the rows modulo B_0 gives $X\bar{Y} = 0$. If the two dimension increments summed to $|H|$, we would have $\operatorname{im}\bar{Y} = \ker\bar{X}$; then $X\bar{Y} = 0$ would say that X annihilates $\ker\bar{X}$. Since $\bar{X}Y = 0$, this would imply $XY = 0$, contradicting $T \neq 0$. The usual rank inequality for a product first bounds the sum of the increments by $|H|$, leaving the bound $|H| - 1$. Adjoining H increases the span of the matrices by exactly $|H| - 1$. The inequality is preserved, and finite induction completes the proof. □

Definition 2.7. The *coordinate components* of a subspace $X \subseteq K^I$ are the connected components of its nonzero coordinate forms $x \mapsto x_i$. Zero coordinate forms are omitted. Equivalently, these are the minimal nonempty blocks G for which $X = \bigoplus_G X[G]$. We call these components *active*: the space X is not identically zero on any coordinate of such a component.

Lemma 2.8. *Let* $\mathbf{1} \in M$, $D = (N : M)$, *and* $E = MD$. *For each active coordinate component* G *of* E, *we have*

$$\dim(M|_G) + \dim D[G] \leqslant \dim E[G] + 1. \tag{*}$$

If $\dim E = \dim D + 1$, *there is such a component with* $\dim(M|_G) = 2$. *If* $E = D$, *then* D *is spanned by common eigenvectors of all the operators* $w \mapsto mw$, $m \in M$.

Proof. Coordinate projections preserving E also preserve D, since

$$m\pi_G(d) = \pi_G(md) \in E \subseteq N.$$

Hence $E[G] = (M|_G)D[G]$. Choose bases in $M|_G$ and $D[G]$. The product coordinates are given by the bilinear forms $(m, d) \mapsto m_i d_i$, whose matrices have rank one. The linear relations among these matrices are exactly the relations among the coordinate forms on $E[G]$, since the products span $E[G]$. Thus the matrix family is connected, and its span has dimension $\dim E[G]$. The first and second factors span spaces of dimensions $\dim(M|_G)$ and $\dim D[G]$: they are the full sets of coordinate forms on the corresponding spaces. Lemma 2.6 gives (*).

The sum of the dimension increments over the components is the total increment. When the total increment is one, (*) gives $\dim(M|_G) \leqslant 2$ for every component. If all these dimensions were one, each m would be constant on each block, because $\mathbf{1} \in M$. Then $ME = E$, so $E \subseteq (N : M) = D$, a contradiction.

When $MD = D$, we have $M \subseteq (D : D)$: every multiplier belongs to the multiplicative stabilizer of D. The restrictions to D of the diagonal operators $w \mapsto mw$ are simultaneously diagonalizable. This can be seen directly: partition I according to equality of all the values m_i. By interpolation, the projection onto each class is a polynomial in finitely many of these diagonal operators, so it preserves D. On each resulting coordinate section all the operators are scalar. Every nonzero vector in that section is therefore a common eigenvector. □

3 Normalizing the algorithm and dimension bounds

3.1 Choosing a hyperplane

For a matrix subspace $H \subseteq \mathrm{Mat}_{p,q}(K)$, set

$$\mathcal{C}(H) = \{a \in K^{p\times 1} : ab \in H \text{ for all } b \in K^{1\times q}\}.$$

This is a space of columns, not a kernel of the matrices in H.

Lemma 3.1. *Suppose an algorithm of format* $\langle p, q, s\rangle$ *and arbitrary length is given over an infinite field, with all* U_i, V_i, W_i *nonzero, where* $p, s \geqslant 1$ *and* $q \geqslant 2$. *Then one can choose a nonzero column* $\eta \in K^{q\times 1}$, *the hyperplane* $F = \{x : x\eta = 0\} \subset Q$, *and nonzero rescalings of the coefficients such that the spaces*

$$M = \{\alpha(\eta u) : u \in P\}, \quad B_F = \{\beta(B) : \operatorname{im} B \subseteq F\}, \quad N = \operatorname{im}\gamma$$

have the following properties:

1. $\mathbf{1} \in M$, *and* M, B_F, N *have full support;*
2. *if* $a \in M$, $b \in \operatorname{im}\beta$, *and* $ab = 0$, *then* $a = 0$ *or* $b = 0$;

3. *for every $G \subseteq I$ and $i \in G$, if $U_i\eta \in \mathcal{C}(\operatorname{span}\{U_j : j \in G\})$, then every column of U_i belongs to this same subspace.*

Here full support means that no coordinate vanishes identically on the entire space.

Proof. Property 3 is obtained by excluding finitely many proper linear subspaces. For fixed G, i, take the inverse image of $\mathcal{C}(\operatorname{span} U_G)$ under $\eta \mapsto U_i\eta$. If this inverse image is proper, choose η outside it.

Also exclude $\ker U_i$ and those η for which $F \subseteq \ker V_i$. The latter condition can occur only when $\operatorname{rk} V_i = 1$ and excludes one line of possible η. This ensures full support for M, B_F. Full support for N follows from $W_i \neq 0$.

For every J with $\operatorname{span} V_J \neq \operatorname{Mat}_{q,s}(K)$, choose a nonzero matrix B_J trace-orthogonal to all V_i, $i \in J$, and require in addition that $B_J\eta \neq 0$. Let $u \neq 0$ and $J = \operatorname{supp}\alpha(\eta u)$. If $\operatorname{span} V_J$ were proper, the algorithm identity would give $B_J\eta u = 0$, a contradiction. Thus V_J spans the whole matrix space. Now if $\alpha(\eta u)\beta(B) = 0$, then $\beta(B)$ vanishes on J, so $B = 0$. This proves property 2.

Over an infinite field, finitely many proper linear subspaces cannot cover the whole space: a product of nonzero linear equations vanishing on them is a nonzero polynomial. Thus a suitable η exists. Next choose u_0 with $c_i = \operatorname{tr}(U_i\eta u_0) \neq 0$ for every i, and replace U_i by $c_i^{-1}U_i$ and W_i by c_iW_i. The algorithm and all ranks are preserved, while $\alpha(\eta u_0) = \mathbf{1}$. All the stated properties survive these rescalings. □

An algorithm equipped with η, u_0 satisfying this lemma will be called *normalized*. Write $m(u) = \alpha(\eta u)$ and set

$$D = (N : M), \qquad E = MD, \qquad Z = (B_F : M), \qquad d = \dim D, \quad e = \dim E, \quad z = \dim Z.$$

We call $g = e - d$ the *dimension increment*. It is nonnegative, since $\mathbf{1} \in M$ implies $D \subseteq E$. For $p = q = s = 3$, we have

$$\dim M = 3, \qquad \dim B_F = 6, \qquad \dim N = 9, \qquad MB_F \perp N.$$

The last orthogonality follows from $B\eta = 0$ and (1.2). Lemma 2.4 immediately gives

$$d + z \geqslant 45 - 2r. \tag{3.1}$$

3.2 Matrix representatives of the conductors

Definition 3.2. For $d_0 \in D$, $z_0 \in Z$, and $u \in P$, define the matrices $C_{d_0}(u)$ and $B_{z_0}(u)$ uniquely by

$$\gamma(C_{d_0}(u)) = m(u)d_0, \qquad \beta(B_{z_0}(u)) = m(u)z_0, \qquad \operatorname{im} B_{z_0}(u) \subseteq F.$$

Existence follows from the definition of the conductor, and uniqueness from injectivity of the coefficient maps. The dependence on each argument is linear.

$$\begin{array}{ccc} P & \xrightarrow{u\mapsto C_{d_0}(u)} & \operatorname{Mat}_{p,s}(K) \\ {\scriptstyle m}\downarrow & & \downarrow{\scriptstyle \gamma} \\ M & \xrightarrow[a\mapsto ad_0]{} & N \end{array} \qquad \begin{array}{ccc} P & \xrightarrow{u\mapsto B_{z_0}(u)} & \{B : \operatorname{im} B \subseteq F\} \\ {\scriptstyle m}\downarrow & & \downarrow{\scriptstyle \beta} \\ M & \xrightarrow[a\mapsto az_0]{} & B_F \end{array}$$

Lemma 3.3. *For all $u, v \in P$, we have*

$$vC_{d_0}(u) = uC_{d_0}(v), \tag{3.2}$$
$$C_{d_0}(u)B_{z_0}(v) = C_{d_0}(v)B_{z_0}(u). \tag{3.3}$$

Let $\mathcal{E} = \gamma^{-1}(E)$. For every $C \in \mathcal{E}$,

$$uCB_{z_0}(v) = vCB_{z_0}(u). \tag{3.4}$$

Moreover, $B_{z_0}(u) = 0$ implies $z_0 = 0$ or $u = 0$.

Proof. Substitute $A = \eta v$ and $\gamma(C) = m(u)d_0$ in (1.2). The right-hand side is symmetric in u, v; varying $B\eta$ gives (3.2). Similarly, the product $m(u)m(v)d_0z_0$ and trace duality give (3.3). For $C = C_{d_0}(w)$, we obtain

$$uC_{d_0}(w)B_{z_0}(v) = wC_{d_0}(u)B_{z_0}(v) = wC_{d_0}(v)B_{z_0}(u) = vC_{d_0}(w)B_{z_0}(u).$$

These C span $\mathcal{E}$, proving (3.4). The last assertion follows from property 2 of Lemma 3.1. □

Lemma 3.4. *For a normalized algorithm of format $\langle p, q, s\rangle$ and arbitrary length r,*

$$ps + (q-1)e \leqslant r.$$

In particular, if $p = q = s = 3$ and $r \leqslant 20$, then $e \leqslant 5$.

Proof. Consider the r linear maps $Q_i : T \mapsto V_iTU_i\eta$ from $\mathrm{Mat}_{s,p}(K)$ to the column space K^q. Identities (1.4) and (3.2) imply

$$\sum_i \gamma(C)_i Q_i(T) = \mathrm{tr}(TC)\eta,$$
$$\sum_i (d_0)_i \alpha(\theta u)_i Q_i(T) = \mathrm{tr}(TC_{d_0}(u))\theta.$$

For the second identity it suffices to take $T = av$ of rank one. By (1.2), its left-hand side is $(vC_{d_0}(u)a)\theta$, as required; these matrices span the whole space. Thus the span of the Q_i contains every map $T \mapsto \mathrm{tr}(TC)\eta$ and every map $T \mapsto \mathrm{tr}(TC)\theta$ with $C \in \mathcal{E}$, $\theta \in K^q$. These two spaces have dimensions ps and qe, and their intersection has dimension e. Their sum therefore has dimension $ps + (q-1)e \leqslant r$. □

Corollary 3.5. *For every field K and integer $n \geqslant 1$,*

$$R_K\langle 2, 2, n\rangle \geqslant \left\lceil \frac{7n}{2} \right\rceil.$$

Proof. An algorithm over K remains valid over an algebraic closure, and deleting zero summands does not increase its length. It therefore suffices to bound the length r of an algorithm with no zero summands over an infinite field. The restriction $r \leqslant 20$ is not assumed here.

Apply Lemma 3.1 with $p = q = 2$ and $s = n$. Injectivity of the coefficient maps gives

$$\dim M = 2, \qquad \dim N = 2n, \qquad \dim B_F = n.$$

Since $\mathbf{1} \in M$, choose x so that $M = \mathrm{span}(\mathbf{1}, x)$. The relation $MB_F \perp N$ and commutativity of the coordinatewise product imply $MN \perp B_F$. Nondegeneracy of the pairing gives

$$\dim MN \leqslant r - n.$$

Consider the surjective linear map

$$\Phi : N \oplus N \longrightarrow MN = N + xN, \qquad (u, v) \longmapsto xu - v.$$

Its kernel consists of the pairs (u, xu) with $u \in N$ and $xu \in N$. This is exactly the condition $u \in D = (N : M)$. By rank-nullity,

$$d = \dim D = 4n - \dim MN \geqslant 5n - r.$$

Since $\mathbf{1} \in M$, we have $D \subseteq MD$, hence $e \geqslant d$. Lemma 3.4 now gives

$$r \geqslant 2n + e \geqslant 2n + d \geqslant 7n - r.$$

Thus $2r \geqslant 7n$, and the integrality of r completes the proof. □

3.3 Joint symmetrizers

Definition 3.6. Suppose $B_z(u) : S \to F$ depends bilinearly on $z \in Z$ and $u \in P$. Its *space of joint symmetrizers* is

$$\Sigma(B) = \{C : P \to S : \ uCB_z(v) = vCB_z(u) \text{ for all } u, v, z\}.$$

In chosen bases, this is the solution space of linear conditions requiring certain matrix products to be symmetric. Lemma 3.3 gives $\mathcal{E} \subseteq \Sigma(B)$.

Lemma 3.7. *If every matrix in a linear space has rank at most one, then the space has either a common one-dimensional column support or a common one-dimensional row support.*

Proof. Choose a nonzero matrix ab. For another nonzero matrix cd, the sum $ab + cd$ has rank two if both a, c and b, d are independent. Thus every element has its first factor proportional to a or its second factor proportional to b. If both types occur with nonproportional free factors, their sum again has rank two. Hence one factor is common to the entire space. □

Lemma 3.8. *Let* $\dim P = \dim S = 3$ *and* $\dim F = 2$*, and suppose that* $B_z(u) = 0$ *implies* $z = 0$ *or* $u = 0$*. If* $\dim Z \geqslant 3$ *and* $\dim \operatorname{span}\{B_z(u) : z, u\} \geqslant 5$*, then* $\dim \Sigma(B) \leqslant 1$*.*

Proof. Let $C \in \Sigma(B)$ be singular, and choose $0 \neq u \in \ker C$. Symmetry implies that all $B_z(u)$ annihilate $\operatorname{im} C$. The map $z \mapsto B_z(u)$ is injective, so

$$\dim Z \leqslant (3 - \operatorname{rk} C) \dim F. \tag{3.5}$$

Consequently, a singular symmetrizer has rank at most one.

Suppose there are an invertible symmetrizer C_0 and a nonzero rank-one symmetrizer C. Choose a row v with $vC \neq 0$ and set $x = (vC)C_0^{-1}$. For $u \in \ker C$, the two symmetry identities give

$$uC_0B_z(x) = xC_0B_z(u) = vCB_z(u) = uCB_z(v) = 0.$$

Thus every $B_z(x)$ annihilates the two-dimensional space $(\ker C)C_0$. Injectivity in z gives $\dim Z \leqslant 2$, a contradiction. Therefore, if C_0 exists, every nonzero element of $\Sigma(B)$ is invertible. There cannot be two independent such elements: the matrix $C_0^{-1}C_1$ has an eigenvalue, making $C_1 - \lambda C_0$ a nonzero singular element.

It remains to consider the case where all symmetrizers have rank at most one. Suppose their space H has dimension at least two. Lemma 3.7 gives two possibilities. If all $C \in H$

have image in a common line $L \subset S$, write two independent elements as $a_1 s, a_2 s$, with $Ks = L$. Symmetry says that $u \mapsto sB_z(u)$ is proportional to each of the independent forms $u \mapsto ua_1$ and $u \mapsto ua_2$. It must therefore vanish: $LB_z(u) = 0$ for all z, u. The span of all $B_z(u)$ then has dimension at most $\dim \mathrm{Hom}(S/L, F) = 4$. If instead all $C \in H$ have a common kernel hyperplane $H_0 \subset P$, then for $x \notin H_0$ the map $C \mapsto xC$ is injective. Its image has dimension $\dim H \geqslant 2$. For a nonzero $u \in H_0$, symmetry forces all $B_z(u)$ to annihilate this image, again giving $\dim Z \leqslant (3 - \dim H)2 \leqslant 2$. Both possibilities are impossible. □

Lemma 3.9. *Under the assumptions of the previous lemma, except for the bounds on* $\dim Z$ *and the span of* B*, suppose that* $Z \neq 0$ *and* $\dim \Sigma(B) \geqslant 5$*. Then there are an invertible matrix* L *and an injective linear map* $f : Z \to F$ *such that*

$$B_z(u) = (Lu^{\mathsf{t}})f(z). \tag{3.6}$$

In other words, its value on a row s *is* $(sLu^{\mathsf{t}})f(z)$*.*

Proof. First establish an auxiliary fact. If independent matrices $A, B \in \mathrm{Mat}_3(K)$ satisfy $Ak = Bk = 0 \Rightarrow k = 0$ for columns k, then the space of matrices T for which both TA and TB are symmetric has dimension at most four. The dual equations are represented by the map

$$(k, l) \longmapsto A[k]_\times + B[l]_\times, \qquad [k]_\times = \begin{pmatrix} 0 & -k_3 & k_2 \\ k_3 & 0 & -k_1 \\ -k_2 & k_1 & 0 \end{pmatrix}.$$

Here k, l are columns, and we write $k \times l = [k]_\times l$ for this formula over K. If the image is zero, multiplication on the right by k and l shows that $A(k \times l) = B(k \times l) = 0$. Hence k, l are dependent. Regard the kernel as a space of 3×2 matrices $(k\ l)$: every matrix in it has rank at most one. If its dimension is at least two, Lemma 3.7 gives a common first or second factor. In the first case, the kernel contains $(k, 0), (0, k)$ with $k \neq 0$, so $A[k]_\times = B[k]_\times = 0$, contrary to the trivial common kernel. In the second case it consists of (ak, bk) with fixed $(a, b) \neq (0, 0)$. The nonzero matrix $aA + bB$ can annihilate $[k]_\times$ only for k on a single line: any nonzero row of that matrix must be proportional to k. Thus the kernel has dimension at most one, and the map has rank at least five. The required space of symmetrizers has codimension at least five. This argument uses the zero diagonal of $[k]_\times$ and does not divide by two.

Choose $z \neq 0$ and represent the two output coordinates of the bilinear map $(u, s) \mapsto sB_z(u)$ by 3×3 matrices A, B. The expression is $(sAu^{\mathsf{t}}, sBu^{\mathsf{t}})$. Since $u \mapsto B_z(u)$ is injective, $Ak = Bk = 0$ implies $k = 0$. The existence of five symmetrizers and the auxiliary fact force A, B to be proportional. Their common nonzero matrix factor L is invertible. Compare it with each coordinate matrix for the other values of z: the common kernel is again trivial, and the same fact forces proportionality to L. The proportionality coefficients define a linear map f. The absence of nonzero pairs with zero product makes f injective. □

Lemma 3.10. *If the family representing the second conductor has form* (3.6)*, then* $\dim NZ = 9z$ *and* $r \geqslant 9z + 3$*. The same conclusion holds if there are row bases* (u_j) *of* P *and* (s_j) *of* S*, and injective maps* $f_j : Z \to F$*, such that*

$$B_z(u_j) = s_j^* f_j(z). \tag{3.7}$$

Here s_j^* *is the corresponding dual-basis column:* $s_k s_j^* = \delta_{kj}$*.*

Proof. Let $z_1, \dots, z_z$ be a basis of Z, and suppose $\sum_a \gamma(C_a) z_a = 0$. Multiply by $m(u)$, use the definition of $B_{z_a}(u)$, and then apply (1.3). This gives

$$\sum_a C_a B_{z_a}(u) = 0 \qquad \text{for every } u.$$

In case (3.6), this is $\sum_a C_a L u^{\mathsf{t}} f(z_a) = 0$. Independence of the rows $f(z_a)$ gives $C_a L u^{\mathsf{t}} = 0$ for all a, u, hence $C_a = 0$. In case (3.7), likewise $C_a s_j^* = 0$ for all a, j, again giving $C_a = 0$. Thus the map $(C_a)_a \mapsto \sum_a \gamma(C_a) z_a$ is injective. Its image is NZ, which consequently has dimension $9z$. Since $Z \subseteq B_F$ and $MB_F \perp N$, we have $M \perp NZ$. Therefore $r \geqslant \dim M + \dim NZ = 3 + 9z$. □

Proposition 3.11. *For a normalized algorithm of format $3 \times 3 \times 3$ and length $r \leqslant 20$, we have*

$$3 \leqslant d \leqslant e \leqslant 5, \qquad z \leqslant 2, \qquad d + z \geqslant 45 - 2r, \qquad e = 5 \Longrightarrow d \geqslant 4, z \leqslant 1.$$

In particular, $r \geqslant 20$; when $r = 20$, the increment $g = e - d$ is zero or one.

Proof. Lemma 2.2, applied to M, Z, gives $\dim MZ \geqslant z + 2$ if $Z \neq 0$. Since $MZ \subseteq B_F$, we get $z \leqslant 4$. Together with (3.1), this gives $d \geqslant 1$. By Lemma 3.4, $e \leqslant 5$.

If $z \geqslant 3$, the span of the matrices $B_z(u)$ has dimension $\dim MZ \geqslant 5$. Lemma 3.8 gives $e \leqslant 1$, so $D = E$ is one-dimensional. Its nonzero vector $w = \gamma(C)$ is a common eigenvector for multiplication by M. Choose an active coordinate i; the space $M_0 = \{m \in M : m_i = 0\}$ has dimension two. For $m(u) \in M_0$, we have $C_w(u) = 0$. Using (3.3) with u_0 satisfying $m(u_0) = \mathbf{1}$ gives $CB_z(u) = 0$. Thus the matrix representatives of $M_0 Z$ lie in $\operatorname{Hom}(S/\operatorname{im} C, F)$, whose dimension is at most four. Lemma 2.2 gives $z+1 \leqslant 4$, hence $z \leqslant 3$. Now $d + z \leqslant 4$, contradicting (3.1). Therefore $z \leqslant 2$ and $d \geqslant 3$.

If $e = 5$ and $z = 2$, Lemmas 3.9 and 3.10 give $r \geqslant 21$, a contradiction. Hence $z \leqslant 1$ and $d \geqslant 4$ in this case. If $r \leqslant 19$, then (3.1) would give $d + z \geqslant 7$; this is impossible when $e \leqslant 4$, while $e = 5$ already gives $d + z \leqslant 6$. Thus $r = 20$. Every remaining pair (d, e) has difference zero or one. □

4 Supports of tests and atomic tests

4.1 Two bounds for tests with prescribed support

Definition 4.1. For $G \subseteq I$, set

$$\mathcal{T}_G = \{C \in \operatorname{Mat}_{p,s}(K) : \operatorname{supp} \gamma(C) \subseteq G\}, \quad \delta_G = \dim \mathcal{T}_G, \quad H_G = \sum_{C \in \mathcal{T}_G} \operatorname{im} C, \quad h_G = \dim H_G.$$

Thus $\mathcal{T}_G$ consists of the tests supported in G, and H_G is their common row support.

Lemma 4.2. *Suppose the columns of all U_i, $i \in G$, lie in a fixed k-dimensional space $H \subset K^{p \times 1}$, with $k < p$, and let $m = |G|$. Then the columns of every $C \in \mathcal{T}_G$ also lie in H, and*

$$\delta_G \leqslant k h_G, \tag{4.1}$$

$$\delta_G + (q-1) h_G \leqslant m, \tag{4.2}$$

$$m + qs + ks \leqslant r + \delta_G. \tag{4.3}$$

Inequality (4.2) requires the normalization $\alpha(\eta u_0) = \mathbf{1}$. In addition,

$$\operatorname{span}\{V_i : i \in G\} \supseteq \{V : \operatorname{im} V \subseteq H_G\}, \qquad \dim \operatorname{span} V_G \geqslant q h_G. \tag{4.4}$$

Proof. For $C \in \mathcal{T}_G$, identities (1.3) give $CB \in \operatorname{span} U_G$ and $AC \in \operatorname{span} V_G$ for all A, B. The first inclusion puts the columns of C in H; the second gives (4.4). Inequality (4.1) follows from the dimension of the space of matrices with columns in H and rows in H_G.

Let $f : K^G \to \mathrm{Mat}_{q,s}(K)$ be given by $f(c) = \sum_{i\in G} c_i V_i$. The injective map $\gamma : \mathcal{T}_G \to K^G$ satisfies $f\gamma(C) = \eta u_0 C$, so $\dim f(\gamma\mathcal{T}_G) \leqslant h_G$. Combining

$$\dim \operatorname{im} f + \dim \gamma\mathcal{T}_G \leqslant m + \dim f(\gamma\mathcal{T}_G)$$

with (4.4) gives (4.2).

For (4.3), restrict the first input by $AH = 0$. All summands indexed by G disappear. On the space of tests with columns in H, restricting γ to $I \setminus G$ has kernel $\mathcal{T}_G$, hence rank $ks - \delta_G$. Choose that many independent coordinate forms. By adding to any test a matrix with columns in H, we can make these forms zero; the product in (1.2) is unchanged because $AH = 0$. We obtain an algorithm for rectangular matrix multiplication with inner dimension $p - k > 0$ and at most $r - m - ks + \delta_G$ summands. Its coefficient map on the space of $q \times s$ matrices is injective, so its length is at least qs. This is (4.3). □

Lemma 4.3. *Bilinear matrix multiplication of format $\langle 3,2,3\rangle$ requires at least fifteen summands.*

Proof. Apply the normalization and conductor construction with $p = s = 3$, $q = 2$. Then $\dim M = 3$, $\dim B_F = 3$, and $\dim N = 9$. If $Z \neq 0$, Lemma 2.2 gives $z \leqslant 1$; if $Z = 0$, the same bound is immediate. Consequently, $d \geqslant 36 - 2r - z \geqslant 35 - 2r$. Lemma 3.4 gives $9 + e \leqslant r$, and $d \leqslant e$. Thus $44 \leqslant 3r$, hence $r \geqslant 15$. Passing to an infinite extension field and deleting zero summands do not increase the length. □

Lemma 4.4. *In an algorithm of format $\langle 3,3,3\rangle$ and length twenty, at most three matrices U_i have their columns in any fixed line. If there are exactly three such indices, then $\delta_G = 1$, and their U_i span the full three-dimensional space of matrices with columns in that line. The same holds for rows and for cyclic permutations of U, V, W.*

Proof. Here $k = 1$. Tests supported in G have a fixed first factor, so $\delta_G = h_G$, and (4.4) gives $3\delta_G \leqslant m$. In the proof of (4.3), deleting G and a further $3 - \delta_G$ terms leaves format $\langle 3,2,3\rangle$. Lemma 4.3 gives

$$15 \leqslant 20 - m - 3 + \delta_G, \qquad m \leqslant \delta_G + 2.$$

Together these bounds yield $\delta_G \leqslant 1$ and $m \leqslant 3$. When $m = 3$, we get $\delta_G = 1$. For nonzero $C = as \in \mathcal{T}_G$, the matrices CB, with B arbitrary, span the whole space $\{ab : b \in K^{1\times 3}\}$ and lie in the span of U_G by (1.3). This proves the assertion about the span. Transposition and cyclic permutation in (1.2) give the other versions. □

Lemma 4.5. *Suppose a linear space of output forms of dimension $\delta \geqslant 3$ is computed from the product of $2 \times n$ and $n \times 2$ matrices. Then the number of summands is at least $3n + \delta - 3$.*

Proof. First let $\delta = 3$. A three-dimensional space of 2×2 test matrices is the trace annihilator of one nonzero matrix. Independent changes of the two bases reduce this matrix to one of two rank cases. In the rank-one case the three computed forms are $xu^{\mathsf{t}}, xv^{\mathsf{t}}, yu^{\mathsf{t}}$, where $x, y, u, v \in K^n$. The first-input forms evaluated on $(0, y)$ separate all y; otherwise the third output form would vanish identically for some nonzero y. Choose n independent such forms and make them zero by a linear substitution $y = Lx$. At most $r - n$ terms remain, computing the first two outputs. They separate all $2n$ coordinates of (u, v), so $r - n \geqslant 2n$.

In the rank-two case the outputs are $xu^{\mathsf{t}} - yv^{\mathsf{t}}, xv^{\mathsf{t}}, yu^{\mathsf{t}}$. Their matrices on $K^n \oplus K^n$ are

$$S_0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad S_1 = \begin{pmatrix} 0 & I \\ 0 & 0 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 & 0 \\ I & 0 \end{pmatrix}.$$

The block map

$$(v_0, v_1, v_2) \longmapsto (S_1 v_2 - S_2 v_1,\ S_2 v_0 - S_0 v_2,\ S_0 v_1 - S_1 v_0)$$

has trivial kernel: use the last two equations to express v_1, v_2 in terms of v_0, and the first becomes $(S_1 S_0^{-1} S_2 - S_2 S_0^{-1} S_1) v_0 = 0$; the operator in parentheses is $-I$. The rank is therefore $6n$. A single bilinear summand contributes the same rank-one matrix to all three S_a, with three scalar coefficients. Its contribution to the block map thus has rank at most that of the corresponding alternating 3×3 matrix, which is at most two. Consequently, $2r \geqslant 6n$. The sign causes no problem in characteristic two.

If $\delta > 3$, the coefficient forms embed the test space injectively into K^r. Choose $\delta - 3$ independent coordinate forms and take their common kernel, of dimension three. On it, $\delta - 3$ summands disappear, and the bound just proved gives $r - (\delta - 3) \geqslant 3n$. □

Lemma 4.6. *In an algorithm of length twenty, at most seven U_i have their columns in any fixed plane $H \subset K^3$. For every set of such indices, $\delta_G \leqslant 2$.*

Proof. First normalize the algorithm using Lemma 3.1; nonzero rescalings preserve the supports and dimensions in question. Lemma 4.2 with $k = 2$ gives $m \leqslant \delta_G + 5$, $\delta_G + 2h_G \leqslant m$, and $\delta_G \leqslant 2h_G$. Hence $h_G \leqslant 2$. If $\delta_G \geqslant 3$, then $h_G = 2$. The tests in $\mathcal{T}_G$ restrict the product of 2×3 and 3×2 matrices and compute δ_G independent output forms. Lemma 4.5 gives $m \geqslant \delta_G + 6$, a contradiction. Thus $\delta_G \leqslant 2$ and $m \leqslant 7$. □

4.2 Atomic tests

Definition 4.7. An *atomic test* is a nonzero rank-one matrix $C = as$, where a is a column and s a row, such that $|\operatorname{supp}\gamma(C)| = 3$. The set $G = \operatorname{supp}\gamma(C)$ is its support; Ka and Ks are its column and row directions.

Lemma 4.8. *For every nonzero test C, we have $|\operatorname{supp}\gamma(C)| \geqslant 3\operatorname{rk} C$. For an atomic test $C = as$ with support G, there are rows b_i and columns c_i such that*

$$U_i = ab_i, \qquad V_i = c_i s, \qquad \gamma(C)_i\, b_j c_i = \delta_{ji} \quad (i, j \in G).$$

In particular, every U_i, V_i on G has rank one. The supports of two atomic tests are either equal or disjoint. In the first case the tests are proportional. The column directions of any three tests with distinct supports are independent, as are their row directions.

Proof. The bilinear form $(A, B) \mapsto \operatorname{tr}(CBA)$ has matrix rank $3\operatorname{rk} C$: after reducing C to rank normal form, it is a sum of $3\operatorname{rk} C$ independent coordinate products. Representation (1.2) contains only $|\operatorname{supp}\gamma(C)|$ rank-one forms, proving the bound.

For $C = as$ and exactly three terms, equality holds in the rank bound. Factoring a full-rank bilinear form through K^3 shows that the first coefficient forms constitute a basis of the forms $A \mapsto bAa$, and the second a basis of the forms $B \mapsto sBc$. Hence $U_i = ab_i$ and $V_i = c_i s$. Equality with the matrix pairing gives the stated weighted duality.

If two atoms share an index, its nonzero U_i fixes their common column direction. Lemma 4.4 permits only three indices in this class, so the supports coincide. The space of tests with this support is one-dimensional, so the tests are proportional. Finally, dependence of the column directions of three distinct atoms would put their nine U_i in one

plane, contrary to Lemma 4.6. For row directions, use the symmetry involving transposition. □

Lemma 4.9. *Let $0 \neq d_0 \in D$, $G = \operatorname{supp} d_0$, and $a = \dim(Md_0)$. Then there is an a-dimensional space of columns H such that*

$$\operatorname{span} U_G = \{T : \textit{all columns of } T \textit{ lie in } H\}.$$

In particular, $\dim \operatorname{span} U_G = 3a$. *If $w \in N \setminus \{0\}$ is a common eigenvector for multiplication by M, then $w = \gamma(C)$ for an atomic test C.*

Proof. Let H be the span of the columns of all $C_{d_0}(u)$. Reciprocity (3.2) gives

$$H = (\ker(u \mapsto C_{d_0}(u)))^{\perp} :$$

a row v annihilates all these columns precisely when $uC_{d_0}(v) = 0$ for every u. Thus $\dim H = \dim Md_0 = a$.

All $C_{d_0}(u)$ are supported in G, so (1.3) implies $H \subseteq \mathcal{C}(\operatorname{span} U_G)$. The equality $\gamma(C_{d_0}(u))_i = (uU_i\eta)(d_0)_i$ shows that $U_i\eta \in H$ for $i \in G$: both sides vanish on the kernel of the family C_{d_0}. Property 3 of Lemma 3.1 now puts every column of U_i in $\mathcal{C}(\operatorname{span} U_G)$. This latter space is itself contained in H, since its elements can be obtained by evaluating matrices in $\operatorname{span} U_G$ on $\eta \neq 0$. Therefore every column of U_i lies in H; the reverse inclusion of the full matrix space is already part of the definition of $\mathcal{C}$.

For a common eigenvector w, we have $w \in D$ and $\dim Mw = 1$. Thus all U_i on its support have one column direction. By Lemma 4.4 the support has at most three indices, and by Lemma 4.8 it has at least three. Its test has rank one and is atomic. □

5 Excluding a positive dimension increment

5.1 Local geometry of a growing component

Lemma 5.1. *If $e = d + 1$, then E has a coordinate component G with the following properties:*

1. $\dim(M|_G) = 2$, $\dim D[G] = 1$, $E[G] = N[G]$, *and* $\dim E[G] = 2$;
2. $6 \leqslant |G| \leqslant 7$, *and the span of U_G is the full space of matrices whose columns lie in a certain plane;*
3. *$\mathcal{T}_G$ is two-dimensional, its column and row supports are two-dimensional, and it contains a rank-two matrix;*
4. *in G, at most two matrices U_i have any fixed column direction, and at most two have any fixed row direction;*
5. *$N[G]$ contains no common eigenvector for multiplication by M.*

Proof. Lemma 2.8 gives a component on which the restriction of M has dimension two. There is a vector $d_0 \in D[G]$ with support exactly G: the supports of $D[G]$ and $E[G]$ coincide, and over an infinite field we can avoid every coordinate hyperplane. By Lemma 4.9, the span of U_G is the full space of matrices whose columns lie in a certain plane. Its dimension is six, so $|G| \geqslant 6$. Lemma 4.6 gives $|G| \leqslant 7$ and $\dim N[G] \leqslant 2$. Now

$$2 + \dim D[G] \leqslant \dim E[G] + 1, \qquad 1 \leqslant \dim D[G], \qquad E[G] \subseteq N[G]$$

force the dimensions to be $1, 2, 2$. If $N[G]$ contained a common eigenvector w, it would lie in $D[G]$ and span it; then $MD[G] = Kw$, contrary to $\dim E[G] = 2$.

The column support of $\mathcal{T}_G$ is two-dimensional by the proof of Lemma 4.9. Inequality (4.2) gives $2 + 2h_G \leqslant 7$, hence $h_G \leqslant 2$. We show that $h_G \neq 1$. If every test has its rows in a line L, then $\mathcal{T}_G = \{as : a \in H\}$ with $\dim H = 2$ and $Ks = L$. Consider the maps

$$f_i : \mathcal{T}_G \to \mathrm{Mat}_{3,3}(K), \qquad f_i(C) = \gamma(C)_i V_i.$$

Their span contains all maps $C \mapsto AC$, which form a space of dimension $3 \dim H = 6$. Projecting the values modulo the space of matrices whose rows lie in L kills these six dimensions. Thus the span of the maps $C \mapsto \gamma(C)_i[V_i]$ has dimension at most $|G| - 6 \leqslant 1$. For indices with $[V_i] \neq 0$, all the nonzero forms $\gamma(\cdot)_i|_{\mathcal{T}_G}$ are consequently proportional: evaluate two proportional maps on any test where one of them is nonzero. There is a nonzero $C \in \mathcal{T}_G$ annihilating all these forms. Its support consists only of indices for which the rows of V_i lie in L. By Lemma 4.4 there are at most three such indices, so C is atomic. For an atom, $U_i = ab_i$; normalization gives $(u_0 a)b_i\eta = 1$ on its support. Hence $m(u)_i = (ua)/(u_0 a)$ is constant there. Thus $\gamma(C)$ is a common eigenvector, which has already been excluded.

Both supports are now two-dimensional. If every test had rank at most one, Lemma 3.7 would make one support one-dimensional. Thus a rank-two test exists.

Three indices with a common column direction would support a nonzero test by Lemma 4.4. This would be an atom and a common eigenvector, a contradiction. For a row direction, apply the same lemma after transposing: three matrices with a common row direction must span the three-dimensional space of all matrices with that direction. But the additional requirement that their columns lie in a plane leaves only two dimensions, a contradiction. □

5.2 Replacing one bilinear pair

Definition 5.2. The *first rank sum* of an algorithm is $\sigma_U = \sum_i \operatorname{rk} U_i$. Among all algorithms of length twenty over the chosen field, choose one minimizing σ_U. A minimum exists because the possible values form a nonempty set of natural numbers. Normalization by nonzero rescalings preserves it. The *bilinear pair* $U \boxtimes V$ denotes the form $(A, B) \mapsto \operatorname{tr}(UA)\operatorname{tr}(VB)$, not the matrix product UV.

Lemma 5.3. *In an algorithm with minimal σ_U, an equality*

$$\sum_i a_i(U_i \boxtimes V_i) = X \boxtimes Y, \qquad a_j \neq 0,$$

implies $\operatorname{rk} U_j \leqslant \operatorname{rk} X$.

Proof. Solve for $U_j \boxtimes V_j$ and substitute into the algorithm. The jth term is replaced by the pair (X, Y) with output $a_j^{-1} W_j$; for $i \neq j$, the output becomes $W_i - a_i a_j^{-1} W_j$. All other first coefficient matrices are unchanged, and the length remains twenty. If $\operatorname{rk} X < \operatorname{rk} U_j$, then σ_U decreases. □

Proposition 5.4. *For an algorithm with minimal first rank sum, the normalization in Lemma 3.1 satisfies $MD = D$.*

Proof. By Proposition 3.11, it suffices to exclude $e = d + 1$. Take the component G from Lemma 5.1. Compress its two outer supports to K^2. The first coefficients have form $U_i = JX_i$, where J is a 3×2 matrix of rank two and X_i are 2×3 matrices. Choose a basis

C, D of the test space such that, on these two supports, C is the identity matrix and D is a nonscalar 2×2 matrix H. We may also require

$$c_i = \gamma(C)_i \neq 0 \quad (i \in G) :$$

each coordinate on $\mathcal{T}_G$ is nonzero, while invertibility and these finitely many conditions are expressed by nonzero polynomials in the two test coordinates. Set $t_i = \gamma(D)_i / c_i$.

Subtracting the two versions of (1.3) for C, D gives a matrix relation with coefficients multiplying the V_i:

$$\sum_{i \in G} c_i (t_i I - H) X_i \boxtimes V_i = 0. \tag{5.1}$$

This equality is interpreted coordinatewise in the matrix of the first factor. It also follows directly from $\eta'(uC) = \sum_i \gamma(C)_i (u U_i \eta') V_i$ for arbitrary u, η'.

By (4.4), $\dim \operatorname{span} V_G \geqslant 6$. If V_G were independent, every first coefficient in (5.1) would vanish. Then the columns of each X_i would lie in an eigenline of H with eigenvalue t_i. A nonscalar 2×2 matrix over an algebraically closed field has at most two eigendirections. By Lemma 5.1, each contains at most two indices, so $|G| \leqslant 4$, a contradiction. Consequently,

$$|G| = 7, \qquad \dim \operatorname{span} V_G = 6.$$

All relations among the V_i are spanned by one nonzero vector w, with $\sum_i w_i V_i = 0$. Each coordinate of the matrix relation (5.1) is a multiple of w. Thus there is a 2×3 matrix T satisfying

$$(t_i I - H) X_i = (w_i / c_i) T \qquad (i \in G). \tag{5.2}$$

We claim that $\operatorname{rk} T = 2$. Let λ, μ be the roots of the characteristic polynomial of H, possibly equal. Both shifts $H - \lambda I$ and $H - \mu I$ are nonzero of rank one, and

$$(H - \lambda I)(H - \mu I) = 0. \tag{5.3}$$

If $T = 0$, the preceding count again gives $|G| \leqslant 4$. Suppose $T = ab$ has rank one. If a is not an eigenvector column of H, then at $t_i = \lambda, \mu$, multiplying (5.2) by the opposite shift gives $w_i = 0$ and puts the columns of X_i in the corresponding eigenline. Away from these two parameters the shift is invertible, and all X_i have the common row direction Kb. Each of these three classes has at most two indices, giving at most six in total, a contradiction.

If $Ha = \lambda a$, multiplying (5.2) by $H - \lambda I$ and using (5.3) shows that, for $t_i \neq \mu$, all columns of X_i lie in Ka. But these indices are exactly the support of the nonzero test $D - \mu C$. There are at most two of them by Lemma 5.1, whereas every nonzero test requires at least three indices. The case $Ha = \mu a$ is analogous. Therefore $\operatorname{rk} T = 2$.

At the roots $t_i = \lambda, \mu$, the left-hand side of (5.2) has rank at most one, so $w_i = 0$. Set $J_0 = \{i \in G : t_i \neq \lambda,\ t_i \neq \mu\}$. For $i \in J_0$, equations (5.3) and (5.2) give

$$c_i (t_i - \lambda) X_i = w_i T + \frac{w_i}{t_i - \mu} (H - \lambda I) T.$$

Pair this with the second bilinear factor V_i and sum. The first term vanishes because $\sum_i w_i V_i = 0$ and $w_i = 0$ outside J_0:

$$\sum_{i \in J_0} c_i (t_i - \lambda)(X_i \boxtimes V_i) = ((H - \lambda I) T) \boxtimes \left(\sum_{i \in J_0} \frac{w_i}{t_i - \mu} V_i \right).$$

Multiplying the first matrices on the left by J transfers the identity to the original U_i. Choose j with $w_j \neq 0$. Then $j \in J_0$, the coefficient of its pair is nonzero, and $\operatorname{rk} U_j \geqslant \operatorname{rk} X_j \geqslant \operatorname{rk} T = 2$ by (5.2). But the first factor on the right, $J(H - \lambda I)T$, has rank at most one. This contradicts Lemma 5.3. □

6 Four atoms: a residual sum of eight terms

6.1 A five-dimensional annihilator

Lemma 6.1. *Let $C_j = a_j s_j$, $1 \leqslant j \leqslant 4$, be rank-one 3×3 matrices, with every three a_j independent and every three s_j independent. Then the space*

$$L = \{T \in \mathrm{Mat}_3(K) : \mathrm{tr}(TC_j) = 0 \ (1 \leqslant j \leqslant 4)\}$$

has dimension five and admits a basis of rank-one matrices.

Proof. Independent changes of basis in the two outer spaces and nonzero rescalings allow the column representatives to be chosen as

$$e_1, \quad e_2, \quad e_3, \quad e_1 + e_2 + e_3.$$

Independently, the row representatives can be reduced to rows with the same coordinates. The annihilator is therefore defined by

$$T_{11} = T_{22} = T_{33} = 0, \qquad \sum_{i \neq j} T_{ij} = 0.$$

These are four independent equations. Consider the six rank-one matrices

$$E_{12} - E_{13},\ E_{21} - E_{23},\ E_{31} - E_{32}, \quad E_{21} - E_{31},\ E_{12} - E_{32},\ E_{13} - E_{23}.$$

They span this hyperplane in the six-dimensional space of off-diagonal coordinates: they are differences of adjacent coordinates along a connected six-cycle. Any five that leave a connected path form a basis. Reversing the changes of basis preserves the ranks. □

Proposition 6.2. *If an algorithm of length twenty has four atomic tests with distinct supports, then there exist a five-dimensional space L, a basis $b_1, \dots, b_5$ of L, eight vectors $w_i \in L$, and matrices $A_{ji} \in \mathrm{Mat}_3(K)$ of rank at most one, satisfying*

$$\sum_{i=1}^{8} \mathrm{tr}(XA_{ji})w_i = \mathrm{tr}(X)b_j \qquad \text{for all } X \in \mathrm{Mat}_3(K), \quad 1 \leqslant j \leqslant 5. \tag{6.1}$$

We call the system (6.1) *a* reconstruction of the outputs $b_1, \dots, b_5$.

Proof. By Lemma 4.8, the four supports are disjoint and contain twelve indices. Let J be the eight remaining indices, and let L be the annihilator of the four tests. These tests satisfy the assumptions of Lemma 6.1; choose its rank-one basis b_j. For $i \in J$, we have $W_i \in L$; set $w_i = W_i$ and $A_{ji} = V_i b_j U_i$. These matrices have rank at most one. In (1.5) with $T = b_j$, the atomic summands vanish: if $U_i = ab_i$ and $V_i = c_i s$, then $V_i b_j U_i = c_i(sb_j a)b_i = 0$, since $sb_j a = \mathrm{tr}(b_j as) = 0$. What remains is (6.1). □

6.2 Coordinates of a three-dimensional kernel

Lemma 6.3. *If* (6.1) *holds, then there exist eight pairwise nonproportional nonzero columns $\mu_i \in K^3$ spanning K^3 and, for each j, a linear map $Z_j : K^3 \to \mathrm{Mat}_3(K)$ such that*

$$A_{ji} = a_{ji} I + Z_j(\mu_i), \qquad (Z_j(x))_{11} = 0.$$

Moreover, $I \in \mathrm{span}_i\, A_{ji}$, and

$$3 \leqslant \dim \mathrm{span}_i\, A_{ji} = 1 + \mathrm{rk}\, Z_j \leqslant 4.$$

Proof. Set $F : K^8 \to L$, $F(c) = \sum_i c_i w_i$. Substituting $X = E_{11}$ in (6.1) shows that each b_j lies in the image of F. Thus F is surjective and $\dim \ker F = 3$. Let μ_i be the coordinates of the restriction of the ith coordinate form to $\ker F$, in a chosen basis of $(\ker F)^*$. They span this entire dual space.

If μ_i, μ_k are dependent, a nonzero form $a\varepsilon_i + b\varepsilon_k$ annihilates $\ker F$ and thus has form $\ell \circ F$ for some $\ell \in L^*$. Applying ℓ to (6.1) gives

$$aA_{ji} + bA_{jk} = \ell(b_j)I.$$

The left-hand side has rank at most two, so $\ell(b_j) = 0$ for every j. Hence $\ell = 0$ and $a = b = 0$, a contradiction. This also excludes zero μ_i.

Set $a_{ji} = (A_{ji})_{11}$. For each matrix entry, the vector of corresponding entries of $A_{ji} - a_{ji}I$ lies in $\ker F$, by (6.1). Thus these matrices arise from the coordinates μ_i through a linear map Z_j; their $(1,1)$ entry is zero.

Choose ℓ with $\ell(b_j) = 1$ and apply it to the reconstruction again: $I = \sum_i \ell(w_i)A_{ji}$. Hence $\operatorname{span}_i A_{ji} = KI \oplus \operatorname{im} Z_j$. This space has dimension at least three: in a basis of matrices of rank at most one, expressing the rank-three identity requires at least three matrices. The upper bound of four follows from $\dim K^3 = 3$. □

6.3 Quadratic separation and concurrent lines

Definition 6.4. A nonzero column $\mu \in K^3$ represents a projective point $[\mu]$; proportional columns represent the same point. A projective line is the set of one-dimensional subspaces of a plane in K^3. Three distinct lines are *concurrent* if they pass through a single point, called their center. For a covering of a family of points by such lines, the *active group* of a line consists of the indices of points on it other than the center.

An index i is *quadratically separable* in the family $[\mu_1], \dots, [\mu_8]$ if there is a homogeneous quadratic polynomial q in three variables with $q(\mu_i) \neq 0$ and $q(\mu_k) = 0$ for $k \neq i$.

Lemma 6.5. *Let the rank-one matrices A_i be pairwise nonproportional, with $I \in \operatorname{span} A_i$, and $\dim \operatorname{span} A_i \leqslant 4$. Then there are an index i and a linear form χ on matrices such that*

$$\chi(I) = 1, \quad \chi(A_i) \neq 0, \quad \chi(A_k) = 0 \ (k \neq i).$$

There is also a homogeneous quadratic form q on matrices, invariant under addition of a scalar multiple of the identity, that separates the same index.

Proof. Partition the family A_i into connected components; their spans form a direct sum. Decompose $I = \sum_H T_H$ along this sum and set $d_H = \dim \operatorname{span} A_H$, $r_H = \operatorname{rk} T_H$. Lemma 2.6 gives $2r_H \leqslant d_H + 1$: the columns of T_H lie in the span of the first factors of A_i, $i \in H$, and its rows in the span of the second factors, so both spans have dimension at least r_H. If $d_H \geqslant 2$, integrality gives $3r_H \leqslant 2d_H$. If $T_H = 0$ whenever $d_H = 1$, we would obtain

$$3 = \operatorname{rk} I \leqslant \sum_H r_H \leqslant \frac{2}{3} \sum_H d_H \leqslant \frac{8}{3},$$

a contradiction. Hence a one-dimensional component has $T_H \neq 0$. Pairwise nonproportionality implies that it contains a single A_i. Projection onto this component, extended to a linear form on all matrices and normalized by its value on I, gives χ.

The matrix $A_i - \chi(A_i)I$ has rank at least two: otherwise $\operatorname{rk} A_i = 1$ would imply $\operatorname{rk} I \leqslant 2$. Choose a nonzero two-by-two minor of this matrix. Taking that minor of $T - \chi(T)I$ defines a quadratic form $q(T)$. For $k \neq i$, it is the corresponding minor of A_k, hence zero. Also $q(T + cI) = q(T)$ because $\chi(I) = 1$. □

Lemma 6.6. *In the model of Lemma 6.3, each j satisfies at least one of the following alternatives:*

1. *there is a quadratically separable index i with $b_j \in Kw_i$;*
2. *the eight points $[\mu_i]$ are covered by three concurrent lines, each containing at least two points other than the center; moreover, b_j lies in the span of the w_i in each of the three active groups.*

Proof. Fix j and suppress it in the notation. If $\dim \operatorname{span} A_i = 4$, the map Z is injective. Thus the A_i are nonzero and pairwise nonproportional: removing the scalar part determined by the $(1,1)$ entry from such a relation would give the same relation among the μ_i. Apply Lemma 6.5. The reconstruction and $\chi(I) = 1$ give

$$b = \sum_k \chi(A_k) w_k = \chi(A_i) w_i.$$

The quadratic form $x \mapsto q(Z(x))$ separates μ_i, since

$$q(Z(\mu_k)) = q(A_k - a_k I) = q(A_k).$$

Now suppose $\dim \operatorname{span} A_i = 3$. Choose a basis of this space consisting of rank-one matrices. A linear combination of them equals I, so both sets of factors are bases and, after absorbing the nonzero coefficients, are dual. Hence a conjugation simultaneously diagonalizes every A_i; each nonzero A_i has exactly one nonzero diagonal coordinate.

After this conjugation, consider

$$\pi(x) = \big((Z(x))_{22} - (Z(x))_{11},\ (Z(x))_{33} - (Z(x))_{11}\big).$$

Its kernel is $\ker Z$: equality of the three diagonal entries means that the matrix is scalar, while the original condition $(Z(x))_{11} = 0$ excludes a nonzero scalar matrix. Thus $\operatorname{rk} \pi = 2$ and $\dim \ker \pi = 1$. Modulo KI, the three coordinate matrix lines give three distinct lines in K^2. Their inverse images under π are three planes through $\ker \pi$, or three concurrent projective lines. They cover all $[\mu_i]$.

Denote the ath diagonal coordinate of the conjugated A_i by c_{ai}. Applying the diagonal coordinate form to the reconstruction gives $b = \sum_i c_{ai} w_i$ for every $a = 1, 2, 3$. Nonzero c_{ai} occur on the corresponding line away from the center. Each active group is nonempty. If all three groups have at least two indices, we obtain the second alternative. Otherwise one group consists of a single index i, and $b \in Kw_i$. The product of linear equations for the other two lines is a quadratic polynomial nonzero at μ_i and zero at every other point. This gives the first alternative. □

Lemma 6.7. *For eight distinct projective points, the following hold.*

1. *At most three indices are quadratically separable.*
2. *If the points are covered by three concurrent lines, each containing at least two points other than the center, then no index is quadratically separable.*
3. *There are at most four distinct centers of such coverings.*

Proof. 1. The evaluation vectors of quadratic forms constitute a subspace of K^8 of dimension at most six. If there are s separable indices, their coordinate vectors lie in this space. On the remaining points, evaluation of quadratic forms has rank at least $\min(3, 8 - s)$: any three distinct projective points can be separated by products of linear forms. Thus $s + \min(3, 8 - s) \leqslant 6$, which implies $s \leqslant 3$.

2. Suppose a quadratic form vanishes at all but possibly one point. Its restriction to a projective line, if nonzero, has at most two zeros. Thus three zeros on a line force its linear equation to be a factor of the form. If the center is not among the eight points, deleting one point leaves seven points on the three lines, in nonempty groups, at least two of which have size at least two. We can choose three, two, and one points on the respective lines. The first triple forces one linear factor, and the second pair forces the other. The last point, outside these two lines, gives a contradiction. If the center is among the eight points and is the point deleted, seven noncentral points remain, at least two on each line, and the same argument applies. If a noncentral point is deleted, the center remains a zero. Two lines still contain at least two noncentral points each; together with the center these give two triples. The form must be the product of equations of these two lines, but a remaining noncentral point on the third line is not a zero of that product.

3. If some line contains at least four points, it must occur in every covering by three lines: otherwise the covering lines would meet it in at most three points. All centers lie on that line. For two distinct centers, at most four points can lie off it, at the four intersections of the remaining pairs of lines; the active-group condition requires at least four. Fix a point off the line. In each covering, its covering line contains another point off the line, with only three possible partners. A partner determines the covering line and its intersection with the original line, hence the center. There are therefore at most three centers.

It remains to consider the case where each line contains at most three points. Every covering has at least two lines containing three points. Two distinct centers cannot use the same such line: otherwise the five points off it would lie in the four intersections of the remaining pairs of lines. Five centers would therefore give ten distinct three-point lines. Each contains three unordered pairs of points, and no pair is repeated. But $10 \cdot 3 > \binom{8}{2} = 28$. □

Lemma 6.8. *In the model* (6.1)*, two independent outputs b, b' cannot have coverings of the second type in Lemma 6.6 with the same center.*

Proof. For a fixed center, each noncentral point determines its line uniquely. Hence the active groups of the two coverings agree up to permutation. Let $E_1, E_2, E_3 \subseteq K^8$ be the coordinate spaces of these three disjoint groups. If $H = F(E_1) \cap F(E_2) \cap F(E_3)$, then $2 \dim H \leqslant \dim \ker F = 3$. Indeed, choose linear maps $s_a : H \to E_a$ with $F(s_a(h)) = h$. The map $(h,k) \mapsto s_1(h) - s_2(h) + s_1(k) - s_3(k)$ is injective and takes values in $\ker F$. Injectivity follows by looking at the second and third disjoint coordinate groups. Thus $\dim H \leqslant 1$, and H cannot contain two independent outputs. □

Theorem 6.9. *Over any field, a reconstruction* (6.1) *of five independent outputs by eight terms of rank at most one is impossible.*

Proof. Construct the common family of eight points using Lemma 6.3. If all five outputs have the first type in Lemma 6.6, then by part 1 of Lemma 6.7 they lie in the span of at most three vectors w_i, contradicting independence. Thus there is at least one covering of the second type. By part 2, there are now no quadratically separable indices, so all five outputs have the second type. Their centers are distinct by Lemma 6.8, contradicting part 3 of Lemma 6.7. □

Corollary 6.10. *An algorithm of length twenty cannot have four atomic tests with distinct supports.*

Proof. Proposition 6.2 reduces such an algorithm to a reconstruction ruled out by Theorem 6.9. □

7 Three atoms and completion of the proof

Proposition 7.1. *For a normalized algorithm of length twenty, the equalities* $MD = D$ *and* $\dim D = 3$ *are impossible.*

Proof. By Lemma 2.8, D is spanned by common eigenvectors for multiplication by M. Choose three independent such vectors. Lemma 4.9 turns them into atomic tests $C_j = a_j s_j$ with distinct supports. By Lemma 4.8, the columns a_j form a basis, as do the rows s_j. Let (u_j) be the row basis dual to (a_j).

Since $\gamma(C_j)$ is a common eigenvector, $C_{\gamma(C_j)}(u)$ is a nonzero constant multiple of $(ua_j)C_j$. Identity (3.3) gives

$$(ua_j)s_j B_z(v) = (va_j)s_j B_z(u).$$

Taking $u = u_k$, we get $s_j B_z(u_k) = 0$ for $j \neq k$. Consequently,

$$B_z(u_k) = s_k^* f_k(z)$$

for certain linear maps $f_k : Z \to F$. Each f_k is injective: a zero would imply $B_z(u_k) = 0$, while $u_k \neq 0$. Lemma 3.10 gives $r \geqslant 9z + 3$. But (3.1) with $d = 3$ and $r = 20$ gives $z \geqslant 2$, hence $r \geqslant 21$, a contradiction. □

Proof of the main theorem. Suppose an algorithm of length $r \leqslant 20$ exists over the original field. Extend the field to an algebraic closure and delete zero summands. Proposition 3.11 shows that the remaining length is twenty.

Among algorithms of this length, choose one with minimal first rank sum, and then normalize it using Lemma 3.1. Proposition 5.4 gives $MD = D$, and Proposition 3.11 gives $d \geqslant 3$. If $d \geqslant 4$, choose four independent common eigenvectors in D. By Lemmas 2.8 and 4.9, they give four atomic tests. Their supports are distinct, since atoms with the same support are proportional. This contradicts Corollary 6.10. Hence $d = 3$, but Proposition 7.1 excludes this case.

Thus no algorithm of length at most twenty exists over the original field either. Therefore $R_K\langle 3,3,3\rangle \geqslant 21$. □